

1. System Architecture & Topology

Next.js 14 Workspace (Landing Page, Customer Portal, Operations Dashboard) paired with React Native / Expo for mobile push alerts.

Python FastAPI microservices running XGBoost/LightGBM risk scoring, Prophet forecasting, SciPy linear programming optimization, and LangChain LLM reporting.

PostgreSQL (relational transactions), Neo4j (multi-tier supplier dependency graph), and MongoDB (unstructured audit logs and LLM reports).

2. Microservices Distribution

- 1. Risk Scoring Service:** FastAPI service training XGBoost classifiers on late delivery, defect rate, and price variance.
- 2. Demand Forecasting Engine:** Prophet time-series forecasting service for quarterly demand requirements.
- 3. What-If & Optimization Engine:** SciPy Optimize / Google OR-Tools linear programming solver.
- 4. LLM Procurement Reporter:** LangChain / LlamaIndex pipeline orchestrating real-time procurement report synthesis.